Gradon Nicholls

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Education

University of Waterloo **Waterloo, ON**
MMath in Statistics, Supervisor: Matthias Schonlau *2022–Expected Dec 2026*

Awards:

- Ontario Graduate Scholarship (OGS): Sep 2023–Aug 2026

Carleton University **Ottawa, ON**
MA in Economics *2016–2019*

University of Waterloo **Waterloo, ON**
BMath in Statistics *2011–2016*

Work

University of Waterloo **Waterloo, ON**
Teaching Assistant *2022–2026*

- Developed and presented tutorial and lecture content.
- Answered student questions on the online discussion board.
- Provided descriptive and actionable feedback while grading assignments and exams.

Various **Waterloo, ON**
Research Assistant *2023–2025*

- **Dr. Leia Minaker** (May 2023–2025)
  - Conducted the cleaning, imputation, and weighting for data from a youth advertising study with complex sampling design.
  - Combined data from multiple sources to construct a dataset containing the number of storeys of buidings in each FSA in Canada.
- **Survey Research Center** (May 2023–2025)
  - Prepared a report on student well-being and food affordability using data from the Student Experience Survey; Logistic regression with variable selection using LASSO.
  - Computed calibration weights for the Waterloo Region Matters Survey to ensure reported statistics are representative of Waterloo Region's demographic groups.
  - Designed and led two workshops for a non-statistician audience:
    - Logistic regression with survey data using SAS software.
    - Introduction to using pre-trained Large Language Models with pytorch.

Independent Contractor **Guelph, ON**
2023

- **Ipsos** (January 2023–August 2023)
  - Proposed methodology for segmentation analysis using only public data sources on businesses.
  - Use google search API and web scraping techniques to build richer dataset.
  - Use kmeans and DBSCAN clustering techniques to identify segments.

Bank of Canada **Ottawa, ON**
Research Assistant (2016–19); Economist (2019–22); Senior Economist (2022) *2016–2022*

- Conducted economic research guiding policy at the currency department.
- Developed computer programs to estimate structural economic models.
- Designed and analyzed surveys of consumers (e.g. Methods-Of-Payments Survey) and businesses (e.g. Merchant Acceptance Survey).
- Contributed to survey methodology: sampling, questionnaire design, weighting, cleaning, imputation.

Employment and Social Development Canada (ESDC) **Gatineau, QC**
Research Assistant *2014–2015*

- Conducted social research focused on people with disabilities.
- Contributed to Statistics Canada publications using survey data from the 2012 Canadian Survey on Disability.
- Presented the research at staff meetings.

University of Waterloo **Waterloo, ON**
Instructional Support Assistant *2013*

- Assisted instructors with the course "Introduction to Computer Science" (CS116).
- Ran weekly tutorials to complement main lectures, using Python to illustrate programming concepts.
- Held office hours to help students with assignments and exams.
- Developed and used programming scripts to test and grade assignments.

Boards / Volunteering / Etc.

2024 Canadian Statistics Student Conference, Memorial University (link) **St John's, NL**
Sessions Committee *2023*

Survey Research Centre **Waterloo, ON**
Student Representative *2022–2023*

ASA DataFest 2023 **Waterloo, ON**
Mentor (Volunteer) *April 2023*

Publications

Journal Articles

[1] Leia M Minaker, Gradon Nicholls, Shruti Dua, Tanya Christidis, Matt Quick, Katherine Perrott, Markus Moos, and Adrian Buttazzoni. Development and Validation of the CanHS: The Canada Housing Stock Datasets for Buildings and Postal Codes. *Urban Transitions*, 2026.

[2] Daniela Balutel, Marie-Hélène Felt, Gradon Nicholls, and Marcel-Cristian Voia. Bitcoin Awareness, Ownership and Use: 2016–20. *Applied Economics*, pages 1–26, 2023.

[3] Kim Huynh, Christopher Henry, Gradon Nicholls, and Mitchell Nicholson. Benchmarking Bitcoin Adoption in Canada: Awareness, Ownership and Usage in 2018. *Ledger*, 5, 2020.

[4] Christopher S Henry, Kim P Huynh, and Gradon Nicholls. Bitcoin Awareness and Usage in Canada: An Update. *The Journal of Investing*, 28(3):21–31, 2019.

[5] Christopher S. Henry, Kim P Huynh, and Gradon Nicholls. Bitcoin awareness and usage in canada. *Journal of Digital Banking*, 2(4):311–337, 2018.

Non-Journal Articles

[6] Heng Chen, Walter Engert, Kim P Huynh, Gradon Nicholls, and Julia Zhu. Cash and COVID-19: The Effects of Lifting Containment Measures on Cash Demand and Use. Staff Discussion Paper 2021-3, Bank of Canada, 2021.

[7] Heng Chen, Walter Engert, Marie-Helene Felt, Kim P Huynh, Gradon Nicholls, Daneal O'Habib, and Julia Zhu. Cash and COVID-19: The Impact of the Second Wave in Canada. Staff Discussion Paper 2021-12, Bank of Canada, 2021.

[8] Kim P Huynh, Gradon Nicholls, and Mitchell Nicholson. 2019 Cash Alternative Survey Results. Staff Discussion Paper 2020-8, Bank of Canada, 2020.

[9] Heng Chen, Walter Engert, Kim P Huynh, Gradon Nicholls, Mitchell Nicholson, and Julia Zhu. Cash and COVID-19: The Impact of the Pandemic on Cash Demand and Use. Staff Discussion Paper 2020-6, Bank of Canada, 2020.

[10] Kim P Huynh, Gradon Nicholls, and Mitchell Nicholson. 2018 Merchant Acceptance Survey. Staff Analytical Note 2019-31, Bank of Canada, 2019.

[11] Kim P Huynh, Oleksandr Scherbakov, and Gradon Nicholls. Explaining the interplay between merchant acceptance and consumer adoption in two-sided markets for payment methods. Staff Working Paper 2019-32, Bank of Canada, 2019.

[12] Walter Engert, Ben Fung, Jozsef Molnar, and Gradon Nicholls. Explaining unusual cash patterns in 2018. Staff Analytical Note 2019-22, Bank of Canada, 2019.

[13] Matthew Till, Tim Leonard, Sebastian Yeung, and Gradon Nicholls. A profile of the labour market experiences of adults with disabilities among Canadians aged 15 years and older, 2012. Technical Report catalogue no. 89-654-X2015005, Statistics Canada, 2015.

[14] Christine Bizier, Matthew Till, and Gradon Nicholls. Learning disabilities among Canadians aged 15 years and older, 2012. Technical Report catalogue no. 89-654-X-No.2014003, Statistics Canada, 2014.

Workshops

Intro. to LLMs with Applications in Text Classification (link; recording) — **Waterloo, ON**
Statistical Consulting and Survey Research Unit — *May 28, 2026*

Analysis of Survey Data: 101 and Beyond (link) — **Waterloo, ON**
Statistical Consulting and Survey Research Unit — *December 4, 2024*

Analysis of Survey Data: 101 and Beyond (link) — **Waterloo, ON**
Statistical Consulting and Survey Research Unit — *December 6, 2023*

Analyzing Surveys with R (link) — **Waterloo, ON**
UW Statistical Workshops and Applications Group (SWAG) — *January 26, 2023*

Research Presentations

Using LLMs to Catch Mistakes in Coding Open-ended Qs — **Saskatoon, SK**
2025 SSC Annual Meeting — *May 25–28, 2025*

Using LLMs to Catch Mistakes in Coding Open-ended Qs (recording) — **Washington, DC**
CIPHER 2025 — *February 26–28, 2025*

Using LLMs to Catch Mistakes in Coding Open-ended Qs — **Waterloo, ON**
Student Seminar Series — *November 14, 2024*

Using LLMs to Catch Mistakes in Coding Open-ended Qs — **Waterloo, ON**
Waterloo Student Conference in Statistics, Actuarial Science and Finance — *October 18–19, 2024*

Using LLMs to Catch Mistakes in Coding Open-ended Qs — **St John's, NL**
2024 SSC Annual Meeting — *June 2–5, 2024*

Measurement and Selection Errors in Recall and Diary Responses — **Virtual**
2021 Joint Statistical Meetings — *August 8–12, 2021*

Payment Acceptance Trends Among Small Merchants in Canada — **Virtual**
ICES VI — *June 14–17, 2021*

Cash and COVID-19: The Effect of the Pandemic... — **Virtual**
AAPOR 76th Annual Conference — *May 11–14, 2021*

Cash Use and Financial Literacy — **Virtual**
BigSurv20 — *November 6–December 4, 2020*

Consumer Payments Behaviour in a Health Crisis (recording) — **Virtual**
Federal Reserve Bank of Atlanta Talk About Payment Webinar — *November 19, 2020*

Cash and COVID-19: The Impact of the Pandemic... — **Virtual**
USC CESR: COVID-19 Work in Progress Conference — *October 29–30, 2020*

2018 Bitcoin Omnibus Survey: Awareness and Usage — **Denver, CO**
2019 Joint Statistical Meetings — *July 27–August 1, 2019*

Financial Literacy and the Demand for Cash — **Warsaw, Poland**
Shaping a New Reality of Cash Conference — *May 29–31, 2019*

Going Cashless at the Point of Sale: Are We There Yet? — **Toronto, ON**
Payments Canada SUMMIT — *May 14–16, 2019*